

Aiden Zhou

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EDUCATION

Yale University

Expected: **May 2028**

Bachelor of Science in **Computer Science** and **Mathematics**

GPA: **4.0/4.0**

- **Relevant Coursework:** Algorithms, Data Structures, Probability Theory, Advanced Linear Algebra, Real Analysis, Discrete Math, Computational Vision, Graph Deep Learning, Fourier Analysis
- **Leadership:** Multimedia Managing Editor of the *Yale Daily News*, Features Editor of the *Yale Scientific Magazine*, Teaching Assistant for Calculus I/II and Real Analysis

EXPERIENCE

Cohere | Incoming Machine Learning Intern

Summer 2026

- Incoming Machine Learning Intern on Cohere's Sovereign AI team, supporting enterprise and government model deployments

Supervised Program for Alignment Research (SPAR) | Research Intern

Feb 2026 – Present

- Developed adaptive benchmarking methods for large language model evaluation, reducing evaluation cost by up to 98% and variance by 50%–70% relative to random item sampling
- Integrated and deployed three Pyro / py-irt response models for ordinal and richer-structure scoring tasks, expanding adaptive evaluation across agentic and safety benchmarks

Wu Tsai Institute at Yale University | Research Intern

Jan 2025 – Oct 2025

- Engineered and evaluated inverse-graphics vision models using PyTorch, Nvdiffrast, and MATLAB
- Built an end-to-end ETL pipeline to clean and regularize 50+ GB of ECoG data from 60+ patients across 85 tasks; ran CUDA/Slurm GPU workflows and cut manual workload by ~5x

Algoverse AI | Machine Learning Researcher

May 2025 – Aug 2025

- Co-authored and presented a paper at MechInterp @ NeurIPS on persistent trends in learning dynamics and feature evolution across 143 checkpoints of Pythia LLMs

PROJECTS

Adaptive Benchmarking for Graded LLM Evaluation (github.com/aidenzhou8/Ordinal-FBM)

- Extended Fluid Benchmarking beyond binary outcomes to support ordinal / continuous IRT methods such as GRM and GPCM, enabling adaptive evaluation for a broader class of benchmarks

NeuronEvolution: Tracking LLM Features over Pre-Training (neuronevolution.streamlit.app)

- Deployed a web app to track feature evolution across 143 checkpoints of Pythia LLMs; implemented Python-based backend and used Snowflake and AWS to ingest model activations into a data warehouse

Data Journalist Portfolio (clippings.me/users/aidenzhou)

- Produced data-driven reporting and interactive graphics for the Yale Daily News on elections, crime, and university governance; several stories were later picked up by national media outlets

SKILLS

- **Languages:** Python, C/C++, Java, JavaScript, MATLAB, SQL, HTML/CSS, Bash, Racket
- **Machine Learning:** PyTorch, PyG, Transformers, CUDA, scikit-learn, Pandas, Pyro, py-irt
- **Systems / Infrastructure:** AWS, Git, GitHub Actions, Docker, Kubernetes, Weights & Biases, Slurm

HONORS AND AWARDS

- **Yale First-Year Summer Research Scholarship in the Sciences and Engineering (2025)**
- **Math & Science Olympiads:** AIME (3x), CMO, UBC Physics Gold (2x)
- **Chess:** US National Master; 4th at the 2022 North American Youth Championship, 34th at the 2022 World Youth Championship, and top-30 finishes at the World Open and North American Open